Real Madrid and recorded VAR-nullified goals in La Liga, 2018-19 to October 2026

Kaggle companion: revised abstract, inclusion rules, data dictionary, reproducible code and sensitivity analyses

Version of 13 October 2026 · Scope: La Liga league matches, eight complete seasons (2018-19 to 2025-26) plus the first eight rounds of 2026-27, up to 12 October 2026 · Code and data: github.com/RoloMagnus/Real_Madrid_VAR_null_goals_outlier_study (folder `kaggle/`). This document supplements, and does not replace, the article draft in `article/`, whose text is unchanged.

Abstract

On 10 October 2026 a widely shared post listed the La Liga clubs with the most goals ruled out by VAR since 2018 and gave Real Madrid 42 (+1 that day). The post does not state its definition, data source or cut-off. We rebuilt event data from four public sources for all 3,118 league matches played in eight complete seasons plus part of 2026-27 and asked whether Real Madrid's recorded count stays elevated after adjusting for exposure and attacking activity, and how much the answer depends on the event definition and on how events are recorded.

Under our inclusion rules we record 50 VAR-nullified goals for Real Madrid and 30 for the next club (Villarreal): 0.160 per team-match against 0.058 for all other clubs (rate ratio 2.76, 95% CI 2.01–3.72). Our reconstruction produces a different count from the post's; because the post's definition and cut-off are not known, we do not regard either count as the correct one. We count a goal that was scored and then ruled out with VAR involvement evident in a public feed. Of Real Madrid's 50 events, 46 carry commentary stating that the goal was given or celebrated and then overturned after a review, 3 show review wording without a clear statement of the initial decision, and 1 lacks explicit VAR wording. The feeds do not allow us to separate a goal overturned by VAR from an on-field disallowance later confirmed by VAR, so we cannot say that the 50 contain only the former.

The adjusted incidence-rate ratio was 2.01 (95% CI 1.45–2.79), compared with an unadjusted ratio of 2.76 (adjusting for offside flags, shots, expected goals, venue and season; 1.84 among top-10 clubs). Across alternative event definitions, sample restrictions, estimators and leave-one-season-out fits the adjusted ratio ranged from 1.72 to 2.34, and Real Madrid had the highest adjusted ratio of 24 clubs with at least 100 matches. The analyses did not detect statistically significant referee heterogeneity ($p = 0.62$) or an additional VAR-specific effect (ratio 1.36, 95% CI 0.82–2.23, $p = 0.23$). These intervals remain compatible with materially different effects, and unequal treatment need not appear as referee-specific concentration. Published VAR accuracy figures do not identify the error rate among Real Madrid's disallowed goals, so we do not use them to rule errors in or out.

Limitations: all events come from public feeds rather than an official VAR log; only Real Madrid's 50 events were checked against video, and comparison clubs' incidents have not been reviewed with the same manual protocol, so a reporting or classification difference cannot be excluded (a protocol and blank audit form are provided). **The elevated recorded rate persists after adjustment, but its cause remains unresolved**: it could reflect tactics not captured by the covariates, reporting or classification differences, model limitations, officiating differences, or a combination.

Key estimates

Quantity	Value	Note
Real Madrid recorded events / league matches	50 / 312	Next club: Villarreal 30
Rate per team-match, Real Madrid vs other clubs	0.160 vs 0.058	Unadjusted rate ratio 2.76 (95% CI 2.01–3.72); exact conditional Poisson test
Adjusted incidence-rate ratio	2.01 (1.45–2.79)	Poisson GLM, SE clustered by match, n = 6,160 team-matches; top-10 clubs 1.84
Range of adjusted ratio in sensitivity analyses	1.72 to 2.34	Sections 5.1 to 5.3
Referee heterogeneity (15 referees)	p = 0.62	Power to detect log-normal rate spread SD 0.5: 46% (section 5.5)
Extra Real Madrid effect at the VAR stage vs on-field stage	1.36 (0.82–2.23)	p = 0.23; compatible with materially different effects

1. Changes in interpretation relative to the draft

The numbers below are unchanged; the wording and the inferences drawn from them are revised as follows.

Draft wording	Revised treatment
The post's exact figures are “not reproducible”; its count is lower than any reconstruction.	“Our reconstruction produces a different count.” The post's definition and cut-off are unknown, so differences are not attributed to error. Section 2.4.
Attack volume “explains about half of the count” (model fitted on other clubs predicts about 25 events).	Removed. Reported instead: adjusted IRR 2.01 (95% CI 1.45–2.79) versus unadjusted 2.76. A forecast from other clubs does not show that attacking activity caused or explained part of the count.
“None of the signatures of targeting”; the rate “does not differ across 15 referees”.	“The analyses did not detect statistically significant referee heterogeneity or an additional VAR-specific effect.” VAR-stage ratio 1.36 (95% CI 0.82–2.23); power of the referee test is limited (section 5.5).
“VAR accuracy (96–98%) is far too high for errors to account for it.”	Removed. Overall accuracy across leagues and decision types does not identify the error rate among Real Madrid's disallowed goals. Title simulations are kept only as sensitivity analyses under assumed error probabilities.
Counting all 50 goals “would change one league title”.	“Restoring all 50 goals while holding subsequent scoring unchanged changes the calculated champion in one season.” Not part of the abstract (section 5.6).
The residual is “most consistent with Real Madrid's attacking profile”.	Cause left open: unmeasured tactics, reporting differences, model limitations, officiating differences, or a combination.
“Nine seasons”.	“Eight complete seasons plus part of 2026-27” (first eight rounds).

2. Explicit inclusion rules

2.1 What “VAR-nullified” means here

A **VAR-nullified goal** is a goal scored by a team in a La Liga league match that was ruled out, where a public feed shows evidence of a VAR review. It is a *recorded-event* definition. Two situations can produce such a record and are not equivalent: (A) a goal awarded on the pitch and then overturned after a VAR review; (B) a goal disallowed on the pitch (flag or whistle) whose disallowance was then checked and confirmed by VAR. The feeds give us no field for the initial on-field decision. We therefore classify each event by the wording of the commentary (evidence tiers, table 2) and treat the strict tier T1 as the closest approximation to situation A; sensitivity analysis D2 uses T1 only. No commentary line among the 394 events contains the words flag, whistle or linesman, but that absence does not show that situation B never occurs in the data. The manual video check of Real Madrid's 50 goals confirmed the goal and the ruling but has no field for the initial on-field decision; the audit protocol in section 6 adds it.

2.2 Inclusion and exclusion rules

Table 1. Rules applied in order. D0 is the primary definition used for all headline estimates.

Rule	Statement	Effect
R1 Competition and period	League matches of La Liga (first division) from 2018-19, the first VAR season, to matches played by 12 October 2026: 3,118 completed matches (8 x 380 + 78 in 2026-27). Cup and European matches are excluded.	3,118 matches; 6,236 team-matches
R2 Candidate event	Flashscore incident feed logs a “Goal Disallowed” for the match, with scorer, minute and reason.	397 incidents
R3 Team	The event is attributed to the team that scored the ruled-out goal, not to the opponent.	–
R4 VAR evidence (any one)	(a) Flashscore commentary from -1 to +3 minutes around the incident carries the VAR icon or VAR, video-assistant, video-review or “disallow” wording (flag `var_flag`); (b) the incident text mentions the video assistant referee; (c) ESPN logs a review-type event within 3 minutes (overturned goal, VAR no-goal decision, “Video Review” delay or VAR decision line).	394 events (primary definition D0); 3 incidents without evidence excluded, none Real Madrid
R5 Goals ruled out on the pitch	Incidents with no VAR evidence are not counted. A separate phrase-based dataset of 298 on-field disallowed goals (a floor) is used only for the VAR-stage comparison.	Excluded from D0
R6 Other VAR decisions	Goals awarded by VAR, penalty and red-card reviews and checks that left the goal standing are not events, because the pipeline starts from disallowed goals.	Not counted
R7 Unit and duplicates	One row per goal. Two ruled-out goals in one match count twice (largest single-match count is 3). Duplicate descriptions of the same goal are merged.	–
R8 Reason	Reason as logged: offside, attacking foul, handball or unspecified video review. Offside player (scorer or build-up teammate) from ESPN events within +-1 minute.	Used in D5 to D7
R9 Boundary case	Second Mbappé goal in the 0-4 Clásico of 26 October 2024 (65'): ruled out by the assistant referee's flag with no VAR review in either feed. Excluded from VAR-nullified goals.	Part of the on-field dataset

Rule	Statement	Effect
R10 Manual validation	All 50 Real Madrid events were checked by the study team against match video (flag `manual_validation`). No comparison-club event has been checked.	Section 6

2.3 Evidence tiers and composition of the 50 events

Table 2. Evidence tier of recorded events, Real Madrid versus all other clubs (primary definition D0). Tiers: **T1** a verdict line (no goal, disallowed, chalked off) plus a line showing the goal was given or celebrated before the review; **T2** VAR wording but the initial decision is unclear (for example only a review-in-progress line); **T3** no explicit VAR text (kept through an ESPN event or a generic “disallow” commentary line). Fisher exact p compares the Real Madrid share with the others.

Characteristic	Real Madrid (of 50)	Other clubs (of 344)	Fisher p
T1 explicit overturn wording	46 (92%)	319 (93%)	0.77
T2 VAR wording, initial decision unclear	3 (6%)	23 (7%)	1.00
T3 no explicit VAR text	1 (2%)	2 (1%)	0.34
ESPN review-type event within 3 min, all seasons	43 (86%)	261 (76%)	0.15
ESPN review-type event within 3 min, 2023-24 onward	18 (86%)	114 (75%)	0.41
Reason: offside	36 (72%)	243 (71%)	1.00
Reason: Foul	6 (12%)	44 (13%)	1.00
Reason: handball	6 (12%)	42 (12%)	1.00
Reason: video review	2 (4%)	15 (4%)	1.00

The evidence mix of Real Madrid's events does not differ detectably from the others (all p above 0.10), which argues against a gross difference in the quality of the feed, but with 50 events it cannot rule out a moderate one. The ESPN review-type event counts any VAR decision line near the minute, including a decision that leaves the goal standing, so it is weak corroboration of an overturn.

2.4 Comparison with the 10 October post

Table 3. Counts in the post and in our reconstruction for events before 10 October 2026. Real Madrid's 50th event (an attacking foul, 71') falls on 10 October. The “difference” is reconstruction minus post and is not an error rate: the post's definition, source and cut-off are not known.

Club	Post	Primary definition (D0)	Offside + foul	Offside only	D0 minus post
Real Madrid	42	49	41	36	+7
Atlético de Madrid	22	26	22	20	+4
Villarreal	22	30	27	23	+8
Celta de Vigo	20	23	20	17	+3
Real Sociedad	20	21	18	14	+1
Sevilla	20	25	19	13	+5
FC Barcelona	19	27	25	23	+8

The offside-plus-foul reading is closest to the post for Real Madrid, Atlético and Celta, but no single definition reproduces all seven clubs. Our reconstruction therefore produces a different count, and the direction of the post (Real Madrid first, by a wide margin) is the only feature we can compare across the two.

3. Data sources and construction

Four public, unofficial sources were queried programmatically and are not redistributed in raw form: **Flashscore** (incidents, live commentary, match statistics; primary source of events and exposure), **ESPN** public API (typed events, box scores, offside events, fouls by zone; independent corroboration), **Transfermarkt** (full referee names) and **Understat** (team-match xG and PPDA). Names were normalised by removing diacritics and token matching with a small alias table; ESPN and Flashscore matches were joined on kick-off hour and score, dropping ambiguous joins. Season coverage is complete: 380 matches in each of eight seasons plus 78 in 2026-27. Table positions are rebuilt from results because no provider supplies them. The provider terms apply, and the repository holds processed, derived tables only. The full list of tables and columns is in appendix A (also `kaggle/data_dictionary.csv`).

Outcome and exposure. The outcome is the number of recorded VAR-nullified goals for a team in a match. The adjusted model is a Poisson GLM with log link: `events ~ Real Madrid + log(1+offsides) + log(1+shots) + xG + home + season fixed effects`, standard errors clustered by match (dispersion 1.03). Offside flags are themselves an officiating outcome, so the estimate is conditional on them; the analysis without them is reported in section 5.2.

4. Reproducible code

Everything in this document is regenerated from the processed tables by four scripts (Python 3.12; pandas, numpy, scipy, statsmodels, matplotlib, reportlab; see `kaggle/requirements.txt`; no network access needed; seed 2026; about 1 minute).

```
python kaggle/reproduce.py        # sensitivity tables -> kaggle/outputs/*.csv, headline.json
python kaggle/make_figures.py     # kaggle/outputs/figures/*.png
python kaggle/data_dictionary.py  # checks that every column is documented -> kaggle/data_dictionary.csv
python kaggle/build_pdf.py        # this document
```

The pipeline that builds `data/processed/` from the public feeds is unchanged (`scripts/01` to `11`; the raw caches, about 1.6 GB, are regenerated by the crawlers). Table 4 compares what `reproduce.py` prints with the values in the article draft.

Table 4. Reproduction check. The Monte-Carlo values differ from the draft by simulation error only.

Statistic	Article draft	reproduce.py
Real Madrid events / next club	50 / 30	50 / 30
Rate per team-match, Real Madrid / others	0.160 / 0.058	0.160 / 0.058
Unadjusted rate ratio (95% CI)	2.76 (2.01–3.72)	2.76 (2.01–3.72)
Adjusted IRR (95% CI)	2.01 (1.45–2.79)	2.01 (1.45–2.79)
Adjusted IRR, top-10 clubs	1.84	1.84
Extra effect at the VAR stage (95% CI)	1.36 (0.82–2.23), p = 0.23	1.36 (0.82–2.23), p = 0.23
Referee homogeneity chi-square, 14 d.f.	11.9, p = 0.62	11.9, p = 0.62
Fixed-score scenario: champion changes	2020-21 only	2020-21
Chance RM champion 2020-21 at assumed p = 1.7 / 5 / 10 / 25 / 50%	1.9 / 5.1 / 9.8 / 24.7 / 49.6%	1.8 / 5.0 / 10.1 / 24.7 / 50.1%

Core of the code (full file: `kaggle/reproduce.py`). Event classification and the adjusted estimator:

```
def classify_events(d):
    """Evidence tiers for each Flashscore 'Goal Disallowed' incident.

    var_confirmed (draft definition) = Flashscore VAR wording OR ESPN review-type event within 3 minutes.
    No feed records the initial on-field decision as a field, so the tiers rest on commentary wording:
      T1 explicit overturn : a verdict line (no goal / disallowed / chalked off) AND a line showing the goal was
                             given or celebrated before the review (celebrations, 'review of the goal', 'changed
                             his mind', 'on the advice of the video assistant referee', 'intervention of the VAR')
      T2 VAR wording, unclear: VAR wording present, but only a review-in-progress line or only a verdict line
      T3 no explicit VAR text: kept only because an ESPN review-type event lies within 3 minutes, or because a
                             generic 'disallowed' commentary line set var_flag (1 Getafe event)
    """
    low = (d.fs_text.fillna("") + " || " + d.var_commentary.fillna("")).str.lower()
    d["text_var"] = low.str.contains(r"\bvar\b|video assistant|video review")  # explicit VAR wording (var_flag is looser)
    d["espn_ok"] = d.espn_corroborated == True  # noqa: E712
    d["var_confirmed"] = d.var_flag | d.text_var | d.espn_ok
    verdict = low.str.contains(r"no goal|disallow|chalked|won.t count")
    given = low.str.contains(r"celebrat|review (?:of )?the \w*\s?goal|review the .{0,20} goal|decided to review|subject to a var review"
                             r"|var review to see|changed his mind|after using var|advice of the video assistant|intervention of the var"
                             r"|signals a var review|legality of the goal|var signal")
    tier = np.where(~d.text_var, "T3 no explicit VAR text",
                    np.where(verdict & given, "T1 explicit overturn wording", "T2 VAR wording, initial decision unclear"))
    d["tier"] = tier
    return d

def fit_irr(t, y, term="is_rm", rhs=ADJ, cluster="fs_id"):
    """Poisson GLM, log link; IRR, 95% CI and p for `term` (cluster-robust SE)."""
    t = t.dropna(subset=["shots", "offsides", "xg", "sot"])
    g = t[cluster].astype("category").cat.codes if isinstance(cluster, str) else cluster(t)
    m = smf.glm(f"{y} ~ {rhs}", t, family=sm.families.Poisson()).fit(cov_type="cluster", cov_kwds={"groups": g})
    b, se = m.params[term], m.bse[term]
    return dict(irr=np.exp(b), lo=np.exp(b - 1.96 * se), hi=np.exp(b + 1.96 * se), p=m.pvalues[term],
                n=len(t), dispersion=m.pearson_chi2 / m.df_resid)
```

5. Sensitivity analyses: overview

The analyses below ask how far the adjusted ratio moves when (i) the event definition changes, (ii) the sample or estimator changes, (iii) the comparison is made club by club, and (iv) recording or classification is assumed to differ between clubs. They are sensitivity analyses of an observational reconstruction, not tests of a mechanism. P-values are descriptive and uncorrected for multiple comparisons.

5.1 Event definition and sample restrictions

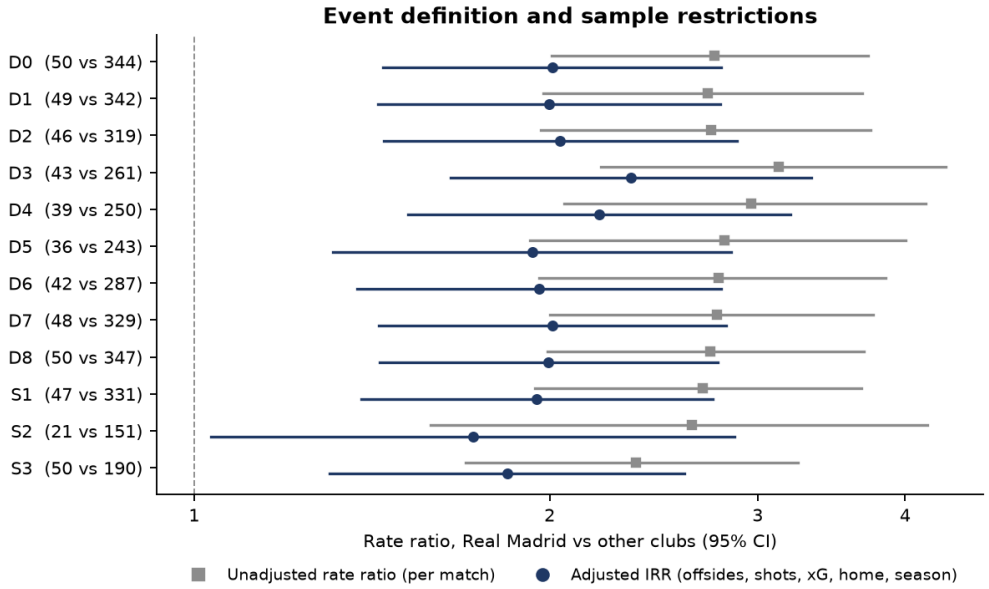


Figure 1. Real Madrid versus other clubs under nine event definitions (D0 to D8) and three sample restrictions (S1 to S3). Labels show Real Madrid events versus other clubs' events.

Definition / restriction	RM	Others	Rate ratio (95% CI)	Adjusted IRR (95% CI)
D0 Primary: VAR evidence in either feed (draft definition)	50	344	2.76 (2.01–3.72)	2.01 (1.45–2.79)
D1 Explicit VAR wording in Flashscore text only	49	342	2.72 (1.97–3.68)	2.00 (1.43–2.79)
D2 Explicit overturn wording (tier T1)	46	319	2.74 (1.96–3.74)	2.04 (1.45–2.88)
D3 ESPN-corroborated only	43	261	3.13 (2.21–4.33)	2.34 (1.65–3.33)
D4 Explicit wording and ESPN-corroborated	39	250	2.96 (2.06–4.16)	2.20 (1.52–3.20)
D5 Offside reason only	36	243	2.81 (1.92–4.00)	1.93 (1.31–2.85)
D6 Offside + attacking foul (the post's apparent scope)	42	287	2.78 (1.96–3.85)	1.96 (1.37–2.79)
D7 Excluding 'unspecified video review'	48	329	2.77 (2.00–3.76)	2.01 (1.43–2.82)
D8 All 397 Goal Disallowed incidents, no VAR requirement	50	347	2.74 (1.99–3.69)	2.00 (1.44–2.77)
S1 Excluding partial 2026-27 season	47	331	2.70 (1.94–3.67)	1.95 (1.38–2.75)
S2 2023-24 onward (ESPN labels overturns explicitly)	21	151	2.64 (1.59–4.18)	1.72 (1.03–2.87)
S3 Top-10 final-table teams only	50	190	2.37 (1.70–3.25)	1.84 (1.30–2.60)

The adjusted ratio ranges from 1.72 to 2.34. Restricting to ESPN-corroborated events raises it (D3 2.34); restricting to 2023-24 onward, where ESPN labels overturned goals explicitly, lowers it to 1.72 with a wide interval (1.03–2.87; 21 Real Madrid events). Every interval stays above 1, but the definitions overlap heavily and are not independent checks.

5.2 Estimator, clustering and influence of single seasons

Estimator for the adjusted Real Madrid IRR	IRR	95% CI
Poisson, SE clustered by match (primary)	2.01	1.45–2.79
Poisson, SE clustered by team-season	2.01	1.53–2.65
Poisson without offsides as a control	2.17	1.55–3.03
Poisson with shots on target instead of shots	2.08	1.50–2.87
Negative binomial (alpha = 0.47)	2.00	1.45–2.77

Clustering by team is not shown: Real Madrid is a single team, so an indicator for it cannot be assessed with team clusters. Clustering by team-season widens the interval to 1.53–2.65, still above 1, although few clusters make that interval approximate. Dropping the offside control does not lower the estimate (2.17), so the adjusted excess does not depend on controlling for an officiating outcome.

Season dropped	RM events left	Adjusted IRR (95% CI)
2018-19	47	2.06 (1.46–2.91)
2019-20	45	1.93 (1.37–2.74)
2020-21	45	2.11 (1.49–3.00)
2021-22	45	2.06 (1.46–2.92)
2022-23	39	1.73 (1.21–2.47)
2023-24	47	2.19 (1.55–3.09)
2024-25	41	2.03 (1.40–2.93)
2025-26	44	2.05 (1.46–2.87)
2026-27	47	1.95 (1.38–2.75)

The estimate is lowest when 2022-23 (39 events left) is dropped (1.73) and highest when 2023-24 is dropped (2.19); no single season produces the excess.

5.3 Comparison clubs

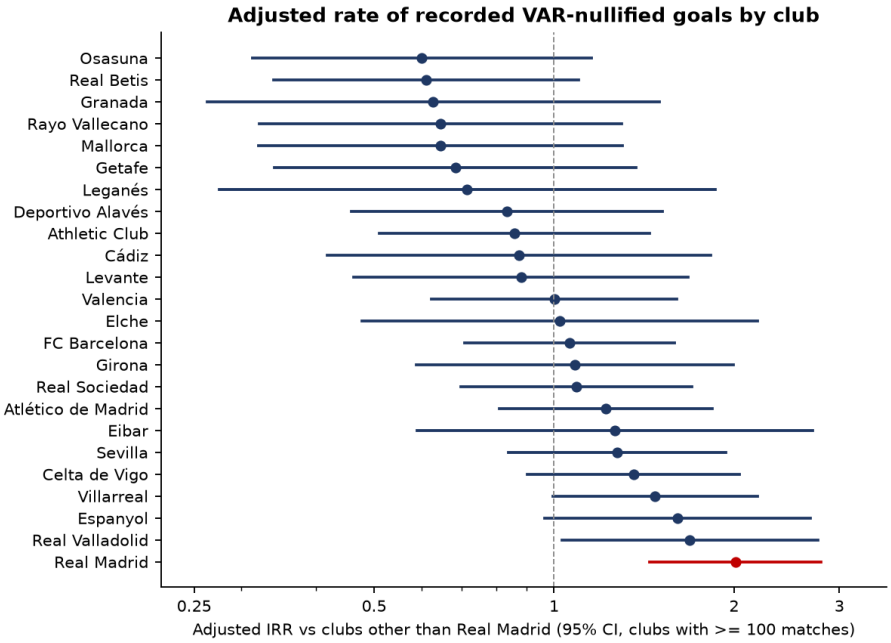


Figure 2. Adjusted IRR of each club with at least 100 league matches, against all clubs other than Real Madrid (same model as the Real Madrid estimate). Real Madrid ranks 1 of 24; the next are Real Valladolid (1.69, 1.03–2.76) and Espanyol (1.61, 0.97–2.68). Intervals for most clubs include 1 and the list contains 24 comparisons, so rank order among the others is imprecise.

5.4 Reporting and classification effects

Reporting or classification would bias the comparison if Real Madrid's incidents were logged or labelled differently from other clubs'. One empirical check and one assumption-based analysis are available without new manual work.

Evidence mix. Table 2 shows the same share of explicit overturn wording (T1: 92% versus 93%), a similar reason mix, and a statistically indistinguishable share with ESPN review-type events (86% versus 76%, p = 0.15). This shows no sign of better recording of Real Madrid's events but is limited by 50 events and by the fact that both providers describe the same broadcasts.

What-if analysis (assumptions, not estimates). Figure 3 and table 5 rescale the adjusted IRR deterministically. If other clubs' true events were c times the recorded ones while Real Madrid's were complete, the ratio would fall to 2.01/c; if k of Real Madrid's 50 events were not true overturns it would fall to 2.01 x (50-k)/50. The factors are illustrative: we have no evidence on their size, which is why the audit in section 6 is needed.

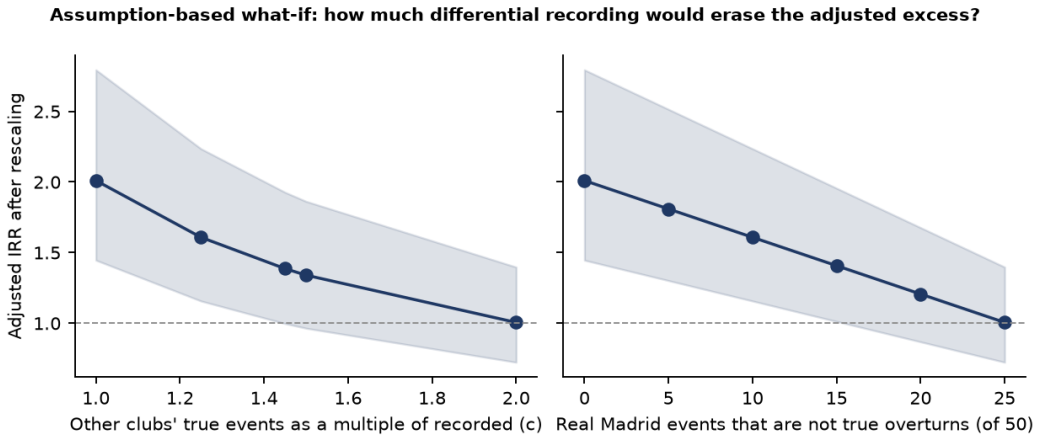


Figure 3. Adjusted IRR (with the 95% CI rescaled by the same factor) under assumed differential recording.

Table 5. Rescaled adjusted IRR under assumed recording differences.

Assumption	Adjusted IRR	95% CI
Other clubs' true events are 1x the recorded ones	2.01	1.45–2.79
Other clubs' true events are 1.25x the recorded ones	1.61	1.16–2.23
Other clubs' true events are 1.45x the recorded ones	1.39	1.00–1.93
Other clubs' true events are 1.5x the recorded ones	1.34	0.96–1.86
Other clubs' true events are 2x the recorded ones	1.00	0.72–1.40
0 of Real Madrid's 50 events are not true overturns	2.01	1.45–2.79
5 of Real Madrid's 50 events are not true overturns	1.81	1.30–2.51
10 of Real Madrid's 50 events are not true overturns	1.61	1.16–2.23
15 of Real Madrid's 50 events are not true overturns	1.41	1.01–1.96
20 of Real Madrid's 50 events are not true overturns	1.21	0.87–1.68
25 of Real Madrid's 50 events are not true overturns	1.00	0.72–1.40

The point estimate would reach 1 only if other clubs' true counts were about twice the recorded counts, or if half of Real Madrid's events were not genuine overturns, while the interval would include 1 from c above about 1.45 or from 16 non-genuine Real Madrid events. These figures show how large a differential would have to be, not that one exists.

5.5 Referee heterogeneity and the VAR stage

Across the 15 referees with at least eight Real Madrid matches (261 matches, 45 events) the chi-square test of a common Real Madrid rate gives 11.9 on 14 d.f. ($p = 0.62$). The analyses did not detect statistically significant referee heterogeneity, but the test has limited power: if true referee rates varied log-normally with a standard deviation of 0.5 it would reject in about 46% of simulated datasets, and with SD 0.25 in 13%. Referee-specific concentration is also not a necessary feature of unequal treatment. Using the same model on stacked VAR-stage and on-field counts, the extra Real Madrid effect at the VAR stage was 1.36 (95% CI 0.82–2.23); the on-field-only adjusted ratio was 1.46. An interval running from a modest deficit to more than a doubling does not support a conclusion of no VAR-specific effect. The on-field dataset is phrase-based, so a floor.

5.6 Title calculation as a fixed-score scenario

Restoring all 50 goals while holding subsequent scoring unchanged changes the calculated champion in one season. The table recomputes each season from all 380 results with La Liga's tie-break order; it describes arithmetic on final scores, not how the matches would have unfolded, and treats every ruling as wrong.

Table 6. Fixed-score scenario: every Real Madrid goal restored, all other scoring unchanged.

Season	RM events	Champion (actual)	RM pts / best other (actual)	RM pts / best other (restored)	Calculated champion changes
2018-19	3	FC Barcelona	68 / 87	69 / 87	No
2019-20	5	Real Madrid	87 / 82	91 / 82	No
2020-21	5	Atlético de Madrid	84 / 86	86 / 86	Yes
2021-22	5	Real Madrid	86 / 73	86 / 73	No
2022-23	11	FC Barcelona	78 / 88	85 / 86	No
2023-24	3	Real Madrid	95 / 85	97 / 85	No
2024-25	9	FC Barcelona	84 / 88	86 / 88	No
2025-26	6	FC Barcelona	86 / 94	88 / 94	No

In 2020-21 Real Madrid (84 points) would level with Atlético (86) at 86 and lead on head-to-head. Because the true error probability is not known for these decisions, the table below varies an **assumed** probability that each ruling was wrong (20,000 simulations per cell); 1.7% is the complement of a multi-league accuracy figure used only as an illustrative low value, and none of the values is a validated estimate for this dataset. Seasons not shown have zero probability at every value.

Table 7. Chance that Real Madrid would be champion in a season it did not win, under assumed independent error probabilities.

Season	p = 1.7%	p = 5%	p = 10%	p = 25%	p = 50%
2020-21	1.8%	5.0%	10.1%	24.7%	50.1%

6. Manual audit protocol for comparison clubs

Status: not yet performed. Only Real Madrid's 50 events were reviewed against video. Reviewing the comparison clubs' incidents with the same protocol would test directly whether part of the difference is a reporting or classification effect, which no analysis of the feeds alone can do. The key and a blank reviewer form are provided (`manual_audit_key.csv`, `manual_audit_form.csv`).

Table 8. Planned audit.

Step	Specification
Scope	All 394 primary-definition events in randomised order (A001 to A394). Priority 1 (181 events): Real Madrid (50) and the five clubs with most events (Villarreal 30, FC Barcelona 27, Atlético de Madrid 26, Sevilla 25, Celta de Vigo 23; 131). Priority 2: the rest.
Blinding	An independent person cuts the clips and keeps the key. Reviewers receive the form (audit id, date, minute) with no club or player names where the footage allows, and are not told the hypothesis.
Fields	initial_onfield_decision (goal given / flag raised before goal / whistle before goal / unclear); var_action (check only / review overturn / review confirmed on-field decision / other); final_decision; final_reason; event_meets_protocol; video source.
Event meets protocol	The ball is seen to enter the goal and the goal is given or flagged by the on-field officials; the final decision is no goal; VAR involvement is visible (review signal, monitor, on-screen graphic or official statement). Subtype A (given then overturned), B (disallowed on field then checked by VAR) or C (other).
Reviewers	Two independent reviewers per priority-1 event, a third to resolve disagreements. Cohen's kappa is reported for event_meets_protocol and for the subtype.
Completeness check	Optional second stage: watch a random sample of matches without a logged event (40 per priority-1 club) to count disallowed goals that the feeds missed, to test for under-recording by club.
Pre-specified analysis	(1) Share of events meeting protocol by group with risk difference and 95% CI; (2) re-estimate the adjusted IRR from audited events only (all subtypes, then subtype A only); (3) report whether the lower bound stays above 1; (4) publish the filled forms, kappa and any deviations.

Table 9. Power of the priority-1 audit to detect a lower share of valid events in one group (normal approximation). A difference of 15 points is detected about half the time; 20 points about 80%.

Share valid, group 1	Share valid, group 2	Real Madrid n	Comparison n	Power (two-sided 5%)
0.95	0.85	50	131	54%
0.95	0.80	50	131	82%
0.95	0.70	50	131	99%

7. Limitations and conclusion

- Events come from public feeds, not an official VAR log; the on-field referee is recorded, not the VAR official. Early seasons are less well corroborated by ESPN.

- The initial on-field decision is not recorded, so situations A and B cannot be separated (section 2.1). The classifier uses commentary wording and is not validated against video.
- Comparison clubs' incidents have not been reviewed manually (section 6). The what-if analysis is arithmetic on assumed factors.
- Offside flags are an officiating outcome. Adjusted estimates are conditional on them; the estimate without them is not lower.
- 50 events over eight complete seasons plus part of 2026-27, with 11 in 2022-23. Tests are descriptive and uncorrected for multiplicity; intervals for team-season clustering are approximate.
- “Top-10” is defined by the final table, a partly post-outcome variable. xG and PPDA are modelled or proxy measures; protocol, technology and coaching changes are absorbed only by season effects.
- Fixed-score scenarios ignore tactical and psychological knock-on effects, and the assumed error probabilities are not validated for these decisions.

Conclusion. The recorded rate of VAR-nullified goals is higher for Real Madrid than for other clubs, and it remains higher after adjusting for offside flags, shots, expected goals, venue and season (adjusted IRR 2.01, 95% CI 1.45–2.79; unadjusted 2.76). The estimate is stable across the definitions, estimators and samples examined here. The analyses did not detect statistically significant referee heterogeneity or an additional VAR-specific effect, but their intervals are wide. **The cause of the elevated recorded rate remains unresolved.** Event-level tracking data, an official VAR log and the manual audit of comparison clubs would address that.

Reference for context only: Spitz J, Wagemans J, Memmert D, Williams AM, Helsen WF. Video assistant referees (VAR): the impact of technology on decision making in association football referees. J Sports Sci 2021;39(2):147-153. doi:10.1080/02640414.2020.1809163. Its accuracy figures cover other competitions and decision types and are not used for inference here.

Appendix A. Data dictionary

Types: string, integer, float, boolean, date/time. Blank values are missing. Club names are normalised across providers. The file `kaggle/data_dictionary.csv` holds the same content and is checked against the actual column headers by `data_dictionary.py`.

data/processed/disallowed_goals_enriched.csv

One row per Flashscore 'Goal Disallowed' incident (397), all clubs, with context. (43 columns)

Column	Type	Description
season	string	Season label, e.g. 2018-19 (August to July; the COVID-delayed 2019-20 finish is mapped explicitly).
fs_id	string	Flashscore match identifier; primary key of the match.
date	date/time	Local match date (Europe/Madrid).
kickoff	date/time	Local kick-off time (Europe/Madrid).
home	string	Home club (normalised name).
away	string	Away club (normalised name).
final	string	Final score as home-away.
team	string	Club whose goal was ruled out.
opponent	string	Opponent of `team` in the match.
venue	string	Home or Away, from the perspective of `team`.
player_short	string	Scorer as abbreviated by Flashscore (e.g. 'Mbappe K.').
player	string	Scorer slug from the Flashscore URL (e.g. 'mbappe-kylian').
minute	string	Minute of the incident as displayed (e.g. 45'+2').
minute_num	integer	Minute as a number; stoppage time is added to the base minute (90+7 = 97).
half	integer	1 or 2, from the base minute; stoppage time inherits the half.
reason_fs	string	Reason logged by Flashscore: offside, Foul (attacking foul), handball or video review (unspecified).
score_home	integer	Running home score before the incident (Flashscore incident feed).
score_away	integer	Running away score before the incident.
fs_text	string	Free text attached to the Flashscore incident, when present (about 78% of rows).
var_commentary	string	Flashscore live-commentary lines from -1 to +3 minutes around the incident that carry the VAR icon, VAR or video wording or the word 'disallow', joined with ' || '. Mostly written as commentary, not an official log.
var_flag	boolean	True if any such commentary line exists or the incident text mentions the video assistant referee. Looser than 'explicit VAR wording': it also fires on a generic 'disallow' line (1 Getafe incident in the 394).
ref_short	string	Referee name as abbreviated by Flashscore (on-field referee, not the VAR official).
pos_before	float	League position of `team` at kick-off, rebuilt from results (points, goal difference, goals); blank in round 1.
pts_before	integer	Points of `team` before the match.
played_before	integer	Matches played by `team` before the match.
opp_pos_before	float	League position of the opponent at kick-off.
team_final_pos	integer	Final league position of `team` in the season (partial table for 2026-27).
opp_final_pos	integer	Final league position of the opponent.
referee_full	string	Full referee name from Transfermarkt match sheet (or majority-vote mapping of the Flashscore name).
referee_source	string	Where `referee_full` came from.
player_full	string	Scorer's full accented name, mapped from the Flashscore slug to ESPN roster names.
context_text	string	Decision sentence extracted from `var_commentary` (or `fs_text`) used for the 'var_decision_text' field of the report.
offside_player_espn	string	ESPN offside event text for the same team within +-1 minute (who was flagged), if any.
category	string	Derived class: offside by goalscorer, offside by teammate in build-up, offside player not identified, foul, handball, other.
half_label	string	1st half or 2nd half.
goal_effect	string	What the goal would have meant given the score before it: Go-ahead / opener, Extend lead, Equaliser, Reduce deficit.
team_gf	integer	Final goals scored by `team` in the match (actual result).

Column	Type	Description
team_ga	integer	Final goals conceded by `team`.
pts_actual	integer	Points `team` earned in the match.
pts_if_allowed	integer	Points if `team` had scored one more goal and nothing else changed (fixed-score what-if).
points_lost	integer	pts_if_allowed minus pts_actual for this incident alone.
result_changed	boolean	True if the fixed-score what-if changes the match result (points_lost > 0).
espn_corroborated	boolean	True if an ESPN review-type event (overturned goal, VAR no-goal decision, or any 'Video Review' / VAR decision line, including 'decision on field stands/cancelled') lies within 3 minutes; blank if no ESPN match record. It does not by itself show that a goal was overturned.

data/processed/team_match.csv

One row per team per league match (6,236 = 3,118 x 2): exposure covariates. (37 columns)

Column	Type	Description
season	string	Season label.
fs_id	string	Flashscore match id.
kickoff	date/time	Local kick-off.
team	string	Club of this row.
opponent	string	Opponent.
venue	string	H or A.
gf	integer	Goals scored by `team`.
ga	integer	Goals conceded.
possession	float	Possession %, Flashscore.
shots	float	Total shots, Flashscore.
sot	float	Shots on target.
corners	float	Corners.
yellow	float	Yellow cards.
shots_off	float	Shots off target.
blocked	float	Blocked shots.
offsides	float	Offside flags against `team` (an officiating outcome; see threats to validity).
fouls	float	Fouls committed.
red	float	Red cards (not listed when zero; set to 0).
xg_fs	float	Flashscore xG; exists from 2022-23 only (55% blank). Used only as a cross-check.
big_chances	float	Flashscore big chances (77% blank).
shots_in_box	float	Shots inside the box (77% blank).
box_touches	float	Touches in the opposition box (77% blank).
pos_before	integer	League position before kick-off.
pts_before	integer	Points before the match.
played_before	integer	Matches played before.
date	date/time	Local match date.
xg	float	Understat expected goals of `team` (1.2% missing); used in the models.
xga	float	Understat expected goals against.
ppda_att	float	Passes allowed by `team`'s opponent in the pressing zone (PPDA numerator component, Understat).
ppda_def	float	Defensive actions by `team` in the pressing zone (PPDA denominator component).
opp_ppda_att	float	Same as ppda_att for the opponent.
opp_ppda_def	float	Same as ppda_def for the opponent.
ppda	float	Passes allowed per defensive action of `team`.
opp_ppda	float	PPDA of the opponent in the same match.
final_pos	integer	Final league position of `team` that season.
top10	boolean	True if `team` finished in the top 10 of that season's final table (post-outcome definition).
pts	integer	Points earned in the match.

data/processed/RM_nullified_goals_breakdown.csv

The 50 Real Madrid goals in the primary definition (28 report fields + manual flag + link). (30 columns)

Column	Type	Description
rank	integer	Chronological order 1-50.
date	date/time	Match date.
season	string	Season (e.g. 2018-2019).
round	float	League round.
team (goal nullified)	string	Always Real Madrid.
rival	string	Opponent.
active_player	string	Scorer.
passive_player	string	Offside teammate in the build-up (ESPN event), when applicable.
local_flag	boolean	True if Real Madrid played at home.
var_reason	string	Offside, Foul, Handball or Video review (unspecified).
context_category	string	Derived class (see disallowed_goals_enriched.category).
minute	integer	Minute (stoppage added).
minute_display	string	Minute as displayed.
half	string	first or second.
score_before_goal (team-opp)	string	Score before the goal, Real Madrid first.
what_the_goal_would_have_been	string	Go-ahead / opener, Extend lead, Equaliser or Reduce deficit.
team_goals_final	integer	Real Madrid final goals.
rival_goals_final	integer	Opponent final goals.
outcome	string	WIN, DRAW or LOSS from Real Madrid's perspective.
points_lost_due_to_nullification	integer	Fixed-score what-if points for this goal alone.
league_position_at_time	float	Real Madrid position at kick-off.
rival_position_at_time	float	Opponent position at kick-off.
points_at_time	integer	Real Madrid points before the match.
referee_on_field	string	On-field referee.
laliga_title_won_flag	boolean	True if Real Madrid won the title that season.
la_liga_champion	string	Champion of the season.
var_decision_text	string	Decision sentence from commentary (blank for 1 event).
var_evidence	string	Label of the evidence class in the report; 'Explicit VAR wording / ESPN review' for all 50 (it does not separate T1/T2/T3; see events_with_evidence_tier).
manual_validation	boolean	True if the study team checked the goal against match video (all 50).
links	string	Video link used in the manual check.

data/processed/onfield_disallowed_events.csv

Goals ruled out on the pitch without VAR wording (298; a floor, phrase-based). (26 columns)

Column	Type	Description
rank	integer	Chronological order.
fs_id	string	Flashscore match id.
date	date/time	Match date.
season	string	Season.
round	integer	League round.
team	string	Club whose goal was ruled out.
rival	string	Opponent.
local_flag	boolean	True if `team` was at home.
active_player	string	Scorer.
reason	string	Offside, Foul, Handball or Other.
kind	string	Commentary pattern that produced the row: flag, goal_disallowed_line, ruled_out, whistle_foul, thought_scored.
minute	integer	Minute (stoppage added).
minute_display	string	Minute as displayed.
half	string	first or second.

Column	Type	Description
score_before_goal	string	Score before the goal, `team` first.
goal_would_have_been	string	Effect of the goal on the score.
team_goals_final	integer	Final goals of `team`.
rival_goals_final	integer	Final goals of the opponent.
outcome	string	WIN, DRAW or LOSS.
points_lost	integer	Fixed-score what-if points.
league_position_at_time	float	Position at kick-off.
rival_position_at_time	float	Opponent position at kick-off.
points_at_time	integer	Points before the match.
referee_on_field	string	Referee.
commentary	string	Source commentary line.
season_final_pos	integer	Final position of `team` in the season.

data/processed/matches.csv

One row per league match (3,118): results and referee (Flashscore). (11 columns)

Column	Type	Description
season	string	Season.
fs_id	string	Flashscore match id.
date_utc	date/time	UTC kick-off.
round	string	Round label.
home	string	Home club.
away	string	Away club.
home_goals	integer	Final home goals.
away_goals	integer	Final away goals.
kickoff_local	date/time	Local kick-off.
date	date/time	Local date.
referee_short	string	Abbreviated on-field referee.

data/processed/final_tables.csv

Final league table per season rebuilt from results (180 rows). (8 columns)

Column	Type	Description
team	string	Club.
pts	integer	Points.
gd	integer	Goal difference.
gf	integer	Goals for.
played	integer	Matches played.
final_pos	integer	Final position (points, goal difference, goals).
season	string	Season.
relegated	boolean	True if relegated.

kaggle/outputs/events_with_evidence_tier.csv

The 394 primary-definition events with the evidence tier added by reproduce.py. (13 columns)

Column	Type	Description
season	string	Season.
fs_id	string	Flashscore match id.
date	date/time	Match date.
team	string	Club whose goal was ruled out.
opponent	string	Opponent.
venue	string	Home or Away.
player_full	string	Scorer.
minute_num	integer	Minute.
reason_fs	string	Reason logged by Flashscore.

Column	Type	Description
tier	string	T1 explicit overturn wording (verdict line plus a line showing the goal was given/celebrated before review); T2 VAR wording but initial decision unclear; T3 no explicit VAR text.
espn_ok	boolean	Same as espn_corroborated, with blank treated as False.
text_var	boolean	True if the Flashscore text contains explicit VAR / video assistant / video review wording.
referee_full	string	On-field referee.

kaggle/outputs/manual_audit_key.csv

Unblinded key for the manual audit: every primary-definition event in randomised order. (11 columns)

Column	Type	Description
audit_id	string	Random-order identifier A001-A394.
priority	integer	1 = Real Madrid or one of the five clubs with most events; 2 = others.
season	string	Season.
fs_id	string	Flashscore match id.
date	date/time	Match date.
team	string	Club.
opponent	string	Opponent.
minute_num	integer	Minute.
player_full	string	Scorer.
reason_fs	string	Reason in the feed.
tier	string	Evidence tier.

kaggle/outputs/manual_audit_form.csv

Blank reviewer form (no club or player names) to be filled from video under the protocol. (15 columns)

Column	Type	Description
audit_id	string	Same id as in the key.
date	date/time	Match date, to locate the video.
minute_num	integer	Minute.
initial_onfield_decision	string	goal_given / flag_raised_before_goal / whistle_before_goal / unclear (reviewer fills).
var_action	string	check_only / review_overturn / review_confirmed_on_field_decision / other.
final_decision	string	no_goal / goal.
final_reason	string	offside / foul / handball / other.
event_meets_protocol	string	Y or N: the incident meets the inclusion rules and the event definition in section 6 of the PDF.
subtype	string	A given then overturned / B disallowed on field then checked by VAR / C other.
reviewer_id	string	Reviewer code.
video_source	string	Clip identifier or URL.
second_reviewer_meets_protocol	string	Independent second reviewer's event_meets_protocol (Y or N).
second_reviewer_subtype	string	Independent second reviewer's subtype.
disagreement_resolution	string	How a disagreement was resolved.
notes	string	Free text.